# In-Class Lab: Lambda Functions in Python

**Basic Exercises:**

**Exercise 1:**

Create a lambda function that adds 15 to a given number and prints the result.

```
add_15 = lambda x: x + 15
print(add_15(10))  # Output: 25
```

**Exercise 2:**

Write a lambda function that multiplies two numbers and prints the result.

**Exercise 3:**

Use a lambda function to find the square of a number.

**Exercise 4:**

Use a lambda function with `filter()` to filter out even numbers from a list.

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
even_numbers = list(filter(lambda x: x % 2 == 0, numbers))
print(even_numbers)  # Output: [2, 4, 6, 8, 10]
```

**Exercise 5:**

Use a lambda function with `map()` to convert a list of temperatures from Celsius to Fahrenheit.

**Exercise 6:**

Write a lambda function to find the maximum of two numbers.

**Exercise 7:**

Use a lambda function with `reduce()` to find the product of all elements in a list.

**Exercise 8:**

Create a lambda function that checks if a number is a multiple of 3.

**Exercise 9:**

Write a lambda function to calculate the area of a rectangle.

**Exercise 10:**

Use a lambda function with `map()` to increment each number in a list by 1.

**Exercise 11:**

Create a lambda function that takes a string and returns its length.

**Exercise 12:**

Write a lambda function that concatenates two strings.

**Exercise 13:**

Use a lambda function with `filter()` to find all numbers greater than 5 in a list.

**Exercise 14:**

Create a lambda function that checks if a string contains a specific letter.

**Exercise 15:**

Write a lambda function to double all elements in a list using `map()`.

**Challenging Exercises:**

**Exercise 16:**

Create a lambda function to check if a given string is a palindrome.

**Exercise 17:**

Write a lambda function to sort a list of tuples based on the second element.

```
tuples = [(1, 'one'), (2, 'two'), (3, 'three')]
```

**Exercise 18:**

Use a lambda function to filter out all words from a list that are longer than 4 characters.

```
words = ["apple", "bat", "cat", "dolphin"]
```

**Exercise 19:**

Create a lambda function to find the factorial of a number using `reduce()`.

```
from functools import reduce
```

**Exercise 20:**

Use a lambda function to sort a list of dictionaries by a specific key.

```
people = [{'name': 'Alice', 'age': 25}, {'name': 'Bob', 'ag
e': 30}, {'name': 'Charlie', 'age': 20}]
```

```
# Output: [{'name': 'Charlie', 'age': 20}, {'name': 'Alice',
'age': 25}, {'name': 'Bob', 'age': 30}]
```

**Exercise 21:**

Write a lambda function to flatten a nested list.

```
nested_list = [[1, 2, 3], [4, 5], [6, 7, 8, 9]]


# Output: [1, 2, 3, 4, 5, 6, 7, 8, 9]
```

**Exercise 22:**

Create a lambda function to count the number of vowels in a string.

**Exercise 23:**

Use a lambda function to find the second largest number in a list.

```
numbers = [1, 2, 3, 4, 5, 6]
```

**Exercise 24:**

Write a lambda function to remove duplicates from a list.

```
numbers = [1, 2, 2, 3, 4, 4, 5]
```

**Exercise 25:**

Create a lambda function to find the longest word in a list of words.

```
words = ["apple", "banana", "cherry", "date"]
```